# Ben Pilger

Systems & Security Engineer | IT Trainer

Stuttgart, Germany +49 160 98641438 ben.pilger@arteii.net
arteii.net github.com/arteiii linkedin.com/in/ben-pilger

## Executive Summary

High-performance Systems and Backend Engineer specializing in **Rust, C++, and low-level Windows Internals**. Track record in launching open-source research initiatives, designing decentralized Zero-Trust architectures, and providing technical leadership for complex enterprise IAM and fraud mitigation services from protocol specification to deployment. Combines advanced offensive security insights with the engineering discipline required to deliver fault-tolerant software at massive scale (>200K RPS) with strict sub-100ms latency guarantees.

## Core Expertise

**Languages:** Rust, C++, C, x86-64 Assembly, Python

**Systems & Security:** Zero-Trust Architecture, PAM, IAM, Cryptography, PKI, Reverse Engineering (IDA Pro, Ghidra), Kernel Development, Debugging & OS Internals (Windows/Linux)

**Infrastructure:** Tokio, Actix, gRPC, REST, TCP/IP, TLS, AWS, Cloudflare, Docker, CI/CD

## Professional Experience

### Freelance Engineering & IT Training
*Stuttgart, Germany*
*Independent Consultant & Technical Trainer*
07/2026 – Ongoing

- **Systems Engineering Consultancy:** Provide specialized research, architectural auditing, and low-level development in Rust, C++, and Assembly, focusing on high-concurrency tooling and security hardening.
- **Trainer:** Available for corporate seminars covering low-level kernel programming (Windows/Linux), networking (TCP/IP), and complex Rust FFI integrations.

### AnyDesk Software GmbH
*Stuttgart, Germany*
*Backend Engineer – Solutions Engineering*
04/2025 – 09/2026

- **Decentralized PAM Architecture:** Pioneered a novel Privileged Access Management solution, conceptualizing an asymmetric key-based configuration protocol and a decentralized PKI-service to eliminate single points of failure.
- **High-Concurrency Infrastructure:** Migrated legacy polyglot services (C++, Erlang, TypeScript) to a high-performance Rust architecture, optimizing systems for >200K RPS and strict sub-100ms latency.

### Freelance R&D / Self-Employed
*Stuttgart, Germany*
*Systems & Security Engineer*
11/2024 – 03/2025

- **Kernel Engineering:** Developed Windows kernel-mode drivers and modules in Rust and C++, backed by deep binary analysis and reverse engineering of core operating system internals.
- **Licensing Platforms:** Architected and deployed scalable user, sales, and license management infrastructures with automated software lifecycle installers and international payment gateways.

### Independent Security Researcher
*Remote*
*Offensive Tooling & Compatibility Engineering*
03/2023 – 08/2024

- **Reverse Engineering:** Bypassed complex runtime integrity protections and anti-tamper mechanisms using static and dynamic analysis tools (IDA Pro, WinDbg).
- **Memory Manipulation & Stability:** Built high-performance user- and kernel-mode tooling for memory injections, maintaining strict runtime stability across hundreds of volatile Windows OS builds.